Miscellaneous Problems

1. Using three yearly moving averages, Determine the trend values from the following data.

Year	Profit	Year	Profit
2001	142	2007	241
2002	148	2008	263
2003	154	2009	280
2004	146	2010	302
2005	157	2011	326
2006	202	2012	353

2. From the following data, calculate the trend values using fourly moving averages.

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998
Sales	506	620	1036	673	588	696	1116	738	663

3. Fit a straight line trend by the method of least squares to the following data.

Year	1980	1981	1982	1983	1984	1985	1986	1987
Sales	50.3	52.7	49.3	57.3	56.8	60.7	62.1	58.7

4. Calculate the Laspeyre's, Paasche's and Fisher's price index number for the following data. Interpret on the data.

Commodities	Base Year		Current Year	
	Price	Quantity	Price	Quantity
A	170	562	72	632
B	192	535	70	756
C	195	639	95	926
D	187	128	92	255
E	185	542	92	632
F	150	217	180	314
7	12.6	12.7	12.5	12.8
8	12.4	12.3	12.6	12.5
9	12.6	12.5	12.3	12.6
10	12.1	12.7	12.5	12.8

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